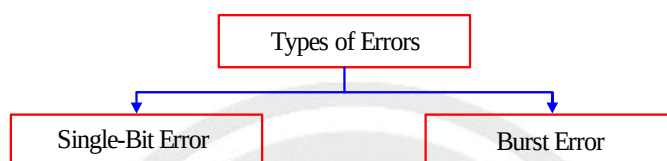


2

ERROR CONTROL

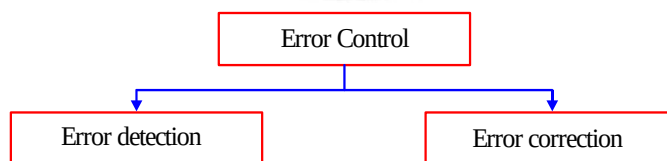
2.1 Error Control



Note:

- The number of corrupted bits or affected bits depends on the data rate and duration of noise.
- The number of corrupted bits or affected bits = Data rate * Noise duration.
- Burst error is more likely to occur than a single bit error.
- Error correction is more difficult than error detection.

Error Detection		Error Correction
1.	Once noticed error simply discard.	Capability of correcting error.
2.	Ask for retransmission.	Does not required retransmission.



1.	Simple Parity	1.	Hamming code
2.	2D parity		
3.	Check sum		
4.	CRC		

2.1.1 Hamming distance

Hamming distance between two Binary string of same size is the number of differences between corresponding bits.
Hamming distance between two Binary string is denoted by $d(x, y)$

$$d(000, 011) = 2$$



$$d(100, 011)=3$$
$$d(10101, 11110)=3$$

Hamming distance can easily be found if we apply XOR operation ($\oplus$) on the two words and count the Number of 1's in the result.

2.1.2 Minimum Hamming distance

In a set of codewords, the minimum Hamming distance is the smallest Hamming distance between all possible pairs of code words.

Valid code word				
0	1	0	(a)	$d(a, b) = 3$
1	0	1	(b)	$d(a, c) = 1$
1	1	0	(c)	$d(a, d) = 2$
0	0	1	(d)	$d(b, c) = 2$
				$d(b, d) = 1$
				$d(c, d) = 3$

Minimum Hamming distance = 1

2.1.3 Minimum Hamming distance for Error detection

To detect 'd' bit error minimum hamming distance required = $d+1$

2.1.4 Min. Hamming distance For Error Correction

To Correct 'd' bit error minimum hamming distance required = $2d+1$

2.2 Simple Parity Check Code

Simple parity

In the Simple parity concept one extra bit (parity bit) is added to each dataword.

Simple parity check can detect all single bit error .

Simple parity check can not detect an even number of errors.

Simple parity check can detect an odd number of errors .

2.3 2D Parity Check Code

2D parity

Two dimensional parity check can detect and correct all single bit error and detect two or three bit error that occur any where in the matrix.

However only some pattern with four or more Errors can be detected.

In a 2D-parity check code, the information bits are organized in a matrix consisting of rows and columns.

For each row and each column one parity check bits is calculated.

2.4 CRC

Length of the dataword = n

Length of the divisor = k

Append $(k-1)$ Zero's to the original message

Perform modulo 2 division

Remainder of division = CRC

Codeword = dataword with Appended $(k-1)$ Zero's + CRC

Note:

1. CRC must be $(k-1)$ bits.
2. If the generator has more than one term and coefficient of x^0 is 1, all single bit error can be detected.
3. If a generator can't divide $x^t + 1$ (t between 0 and $n - 1$) then all isolated Double error can be detected.
4. The generator that contains a Factor of $x + 1$ can detect all odd numbered errors.

2.4.1 A good polynomial generator needs to have the following characteristics

1. It should have at least two terms.
2. The coefficient of the term x^0 should be 1.
3. It should not divide $x^t + 1$, for t between 2 and $n - 1$.
4. It should have the factor $x + 1$.

2.5 Hamming Code

1. Hamming code is used for error correction.
 2. Hamming code can correct 1 bit error only.
 3. Hamming code can detect upto 2 bit error.
- m = Message bits
 r = redundant bits or Check bits or parity bits or extra bits
 $n = m + r$ (n = codeword)

According to the hamming code, number of redundant bits

$$m + r + 1 \leq 2^r$$

where r = lower limit

